Incentive-based Fleet Rebalancing in Vehicle-Sharing Systems via Equilibrium-Constrained Optimal Control

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***Abstract*— We propose an incentive-based fleet rebalancing scheme for vehicle-sharing systems that uses fare discounts to steer users toward alternative pick-up locations. The discounts solve an optimal control problem that trades off users leaving the service against incentive costs over a day of operation. Users' responses to incentives are modeled as a Nash equilibrium of logit models coupled through shared vehicle-availability constraints, which makes the control problem a mathematical program with nested equilibrium constraints. We develop a first-order algorithm that backpropagates through the equilibria via nonsmooth implicit differentiation and enjoys convergence guarantees. In a case study that uses Helsinki bike-sharing data, our scheme reduces the number of leaving users up to 77% relative to no incentives, and outperform heuristic policies while burning up to 46% less incentive budget.**

I. Introduction

Vehicle-sharing services, such as car- and bike-sharing, are increasingly promoted as a sustainable alternative to private vehicles in cities [1]. In new-generation one-way, instant-access, and free-floating systems, users pick up vehicles without reservation and drop them off anywhere in the operational area [1], [2]. However, since trip demand is asymmetric in space and time, vehicles accumulate in some areas and become scarce in others [1], [3]. As users are reluctant to walk far to reach a vehicle [3], [4], such shortages leave demand unserved, pushing users to other transportation modes and suppressing future demand [1]. Keeping the fleet balanced is therefore central to their success.

Fleet rebalancing is either operator-based or user-based [2]. In operator-based relocation, the operator repositions vehicles directly, via staff or, in autonomous fleets, by commanding the vehicles, with decisions computed, e.g., via mixed-integer programming [5], fluid models [6], or approximate dynamic programming [7]. While effective, operator-based relocation is costly and limited by the available staff [3], [8]. In user-based relocation, the operator instead offers incentives, such as fare discounts, steering users toward pick-up or drop-off locations that rebalance the fleet [3], [8]. Surveys show that many users accept moderate detours for discounts [3], [4], and incentives can substantially reduce the staff needed for rebalancing [9].

Designing incentives requires predicting users' responses, for which logit models [10] are the standard tool in vehicle sharing [3], [8], [9], [11]. Incentives have been designed per user request, trading off incentive costs against demand loss predicted via Markov chains [9], or via approximate dynamic programming policies that learn user preferences online [11]. Other works take a system-level view, either jointly optimizing alternative destinations and fare discounts for all vehicles in use [12], or setting incentives per station and time of day within a time-space network of the fleet dynamics [8]. However, these approaches predict each user's choice independently of the others [12]. In [8], demand exceeding the available vehicles is rationed only afterward, so users who find no vehicle are lost rather than redirected, and incentives are optimized period by period via simulated annealing. In practice, users in different areas compete for the same vehicles, and neglecting this may yield incentives that attract more users to an area than it has vehicles, wasting budget [8]. Such competition among users has been captured via dynamic user equilibrium models [13], but only to evaluate predefined incentive strategies.

Optimizing incentives against such equilibrium responses results instead in a mathematical program with equilibrium constraints (MPEC) [14]. Over a day of operation, the equilibria are nested through the fleet dynamics, yielding an optimal control problem with equilibrium constraints (OCPEC) [15]. Numerical methods for OCPECs mostly replace each equilibrium with its complementarity conditions and solve the resulting problem as a single MPEC, via relaxation [16] or Newton-type methods [15]. When each equilibrium is a single-valued Lipschitz function of its parameters, one can instead compute the chain forward and differentiate it backward in time, similar to closed-loop optimization of MPC [17]. For games, however, differentiating through equilibria has been limited to static interventions [18], [19].

In short, incentive schemes for vehicle sharing either neglect the competition among users or do not optimize over it, and, to the best of our knowledge, no method optimizes a finite-horizon cost through a chain of equilibria among competing users with convergence guarantees. This paper closes this gap by following the second computational route described above. Our contributions are threefold.

- We model the users' response to incentives as a generalized game in which the constrained logit models [20] of users located in different regions are coupled by the shared availability of vehicles, capturing the competition for vehicles that existing incentive schemes neglect.
- We cast incentive-based rebalancing as an OCPEC, in which the incentives drive the fleet through the users' equilibrium response, and develop a first-order algorithm that backpropagates through the chain of equilibria to evaluate gradient information. Our approach

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builds on the algorithm in [17], and extend it to game-theoretic settings with entropic lower-level costs.
- We validate the proposed OCPEC approach in a case study on Helsinki city-bike data [21], comparing the optimized incentives against heuristic and myopic policies, in terms of users leaving the service and incentive budget spent. Code and data are publicly available[1].

A. Notation and preliminaries

We denote by $\mathbb{N}$ the nonnegative integers and by $\mathbb{R}_+$ ($\mathbb{R}_{>0}$) the nonnegative (positive) reals. $\mathbb{1}_N$ and 0_N are the all-ones and all-zeros vectors in $\mathbb{R}^N$, $0_{N\times M}$ the zero matrix, I_N the identity, and $e_t \in \mathbb{R}^T$ the t-th canonical basis vector; dimensions are omitted when clear. Subscripts denote time, superscripts indices, and superscripts in parentheses iterations. $(v_1, \dots, v_n)$ and $(v_a)_{a\in\mathcal{A}}$ stack vectors or matrices vertically, the latter following the order of a finite ordered set $\mathcal{A}$. $\operatorname{diag}(v)$ is the diagonal matrix of v and $\otimes$ the Kronecker product; vector inequalities, $\log(\cdot)$, $\exp(\cdot)$, and $\max(0,\cdot)$ act entry-wise. $\mathbb{S} \coloneqq \{p \in \mathbb{R}^n_+ \mid \mathbb{1}^\top p = 1\}$ is the probability simplex and $\mathbb{S}^N$ its N-fold product. For a nonempty closed convex set $\mathcal{K} \subseteq \mathbb{R}^n$, $\Pi_{\mathcal{K}}(v) \coloneqq \arg\min_{y\in\mathcal{K}} \|y - v\|_2$ is the projection onto $\mathcal{K}$ and $\mathcal{N}_{\mathcal{K}}(v) \coloneqq \{z \mid z^\top(y - v) \le 0,\ \forall y \in \mathcal{K}\}$ its normal cone at $v \in \mathcal{K}$. To differentiate nonsmooth maps, we use the following generalization of the Jacobian, where $\rightrightarrows$ denotes set-valued maps.

Definition 1 (Conservative Jacobian [22])**.** Given a locally Lipschitz continuous function $f : \mathbb{R}^n \to \mathbb{R}^m$, we say that $\mathcal{J}_x f : \mathbb{R}^n \rightrightarrows \mathbb{R}^{m\times n}$ is a conservative Jacobian for f if $\mathcal{J}_x f(x)$ has a closed graph, is locally bounded, and is nonempty with

$$\frac{d}{dt} f(\varphi(t)) = \mathcal{J}_x f(\varphi(t)) \cdot \dot{\varphi}(t), \tag{1}$$

where $\varphi : \mathbb{R} \to \mathbb{R}^n$ is an absolutely continuous curve. A locally Lipschitz function that admits a conservative Jacobian is called path-differentiable.

II. Modeling the vehicle-sharing system

This section introduces the mathematical models of the vehicle-sharing system, describing how users influence the distribution of vehicles and user behavior, capturing how users respond to the incentives offered by the operator. Finally, connecting these two, we formally define the incentive-response map.

A. vehicle-sharing system dynamics

We consider a free-floating vehicle-sharing system where vehicles can be parked anywhere within an operational area divided into regions [2]. We label these using $i \in \mathcal{R}$, where $\mathcal{R} \subset \mathbb{N}$ denotes the set of all regions and $N = |\mathcal{R}|$ is the number of regions. We assume large parking capacity, so that any amount of vehicles can occupy a region at any time.

We assume the vehicle-sharing system to be instant access, namely, users can take available vehicles at any time and access them instantly, without prior reservation [2]. We divide time into intervals of length Δt, matching the time resolution of the user demand forecasts, and we describe users and vehicles in terms of flows that are constant within each interval. We denote by $x \in \mathbb{R}^N_+$ the vehicle distribution, whose entry $x^i \in \mathbb{R}_+$ represents the number of vehicles available in region i at the beginning of an interval. Moreover, we assume an accurate forecast of future user demand flows $d^{\mathrm{pu},i}, d^{\mathrm{do},i} \in \mathbb{R}_+$, defined as the number of users per minute located at region i that need to pick up a vehicle, and the number of users per minute that need to drop off at region i, respectively. We collect these user demand flows in the vectors $d^{\mathrm{pu}} \coloneqq (d^{\mathrm{pu},i})_{i\in\mathcal{R}}$ and $d^{\mathrm{do}} \coloneqq (d^{\mathrm{do},i})_{i\in\mathcal{R}}$, both in $\mathbb{R}^N_+$.

The operator can influence users' choice of pick-up location by offering incentives in the form of fare discounts ranging from 0% to 100% [3]. These incentives $\theta^i \in [0,1]$, for all $i \in \mathcal{R}$, are broadcast to all users in real time via the vehicle-sharing platform, and if a user picks up a vehicle from region i, they can redeem the corresponding discount. We collect these incentives in the vector $\theta \coloneqq (\theta^i)_{i\in\mathcal{R}} \in [0,1]^N$.

Due to these fare discounts, users located in region j may decide to pick up their vehicle at another region $i \in \mathcal{R}$ instead. We denote as $u^{\mathrm{pu},i} \in \mathbb{R}_+$ the number of users per minute picking up vehicles at region i, and refer to this as the users pick-up flow at region i. We collect the users' pick-up flows across all regions in the vector $u^{\mathrm{pu}} \coloneqq (u^{\mathrm{pu},i})_{i\in\mathcal{R}} \in \mathbb{R}^N_+$.

Users' drop-off d^{do} and pick-up u^{pu} flows affect the vehicle distribution across regions, according to the dynamics

$$x_{t+1} = x_t + \Delta t \left(d^{\mathrm{do}}_t - u^{\mathrm{pu}}_t\right), \quad (t \in \mathbb{N}) \tag{2}$$

where the subscript t indicates the time interval. Note that (2) is the exact discretization of the continuous-time conservation law $\dot{x} = d^{\mathrm{do}} - u^{\mathrm{pu}}$, since the flows are constant within intervals. Moreover, since users can only pick up vehicles that are available, the pick-up flow must at all times satisfy

$$u^{\mathrm{pu}}_t \le x_t/\Delta t, \quad (\forall t \in \mathbb{N}) \tag{3}$$

which also ensure positive vehicles distributions $x_{t+1} \ge 0$.

B. Modeling user behavior

This section describes how the fare discounts θ and the vehicle availability x determine the user behavior within each time interval[2]. We build our model in two progressive steps: (a) first, we consider the users located in a single region in isolation, and model their choices with a logit model constrained by the availability of vehicles [20]; (b) then, we extend the model to account for users across all regions, who compete to access the same vehicles, by means of a generalized game.

a) Single region: We first consider the users located in a single region, say $j \in \mathcal{R}$, who request vehicles at rate $d^{\mathrm{pu},j}$. Each of them either picks up a vehicle in some region $i \in \mathcal{R}$, possibly different from j, or leaves the service, e.g.,

[1] https://github.com/tihanyidani/vehicle-sharing

[2] hence, we omit the time index t.

in favor of another transportation mode. We collect these options in the choice set $\mathcal{C} = \mathcal{R} \cup \{\emptyset\}$, where $\emptyset$ denotes leaving. We collect the probabilities of these choices in $p^j \in \mathbb{R}^{|\mathcal{C}|}$, indexed by $\mathcal{C}$ with the regions first and the leaving option $\emptyset$ last, so that $p^{j,c}$ is the probability of choice c. These probabilities follow the constrained logit model [20]

$$p^j(x,\theta) = \arg\max_{y \in \mathbb{S}} \; y^\top r^j(\theta) - y^\top \log(y) \tag{4a}$$

$$\text{s.t.} \quad d^{\mathrm{pu},j}\, \Xi\, y \le x/\Delta t. \tag{4b}$$

where $\mathbb{S} \coloneqq \left\{p \in \mathbb{R}^{|\mathcal{C}|} \mid p \ge 0,\ \mathbb{1}^\top p = 1\right\}$ is the probability simplex, $\log(\cdot)$ is applied entry-wise, $\Xi \coloneqq \begin{bmatrix} I_N & 0_{N\times 1}\end{bmatrix}$ is a selection matrix removing the leaving component, and $r^j(\theta) \in \mathbb{R}^{|\mathcal{C}|}$ models the incentive-dependent rewards associated with each choice, namely, for all $c \in \mathcal{C}$,

$$r^{j,c}(\theta) \coloneqq \begin{cases} \lambda^{j,c}\,\theta^c - w^{j,c}, & \text{if} \quad c \neq \emptyset, \\ -w^{j,\emptyset}, & \text{if} \quad c = \emptyset. \end{cases} \tag{5}$$

with $\lambda^j \in \mathbb{R}^N_+$ modelling the sensitivity of the users to the incentives in each region, $w^{j,i} \in \mathbb{R}_+$ the base cost[3] of picking up a vehicle in region i for users located in region j, and $w^{j,\emptyset} \in \mathbb{R}_+$ is the base cost of leaving the service, with $w^j \in \mathbb{R}^{|\mathcal{C}|}_+$. For a large number of users, $p^{j,i}$ is also the fraction of users choosing i. Hence, $d^{\mathrm{pu},j}\,\Xi\,p^j$ is the pick-up flow generated by these users, and the constraint (4b) enforces the capacity constraints[4] (3).

b) Multi-region: In practice, users access the service from all regions simultaneously, and rank the same pick-up choices differently depending on their specific location[5]. Thus, we divide users in *user types*, where user type $j \in \mathcal{R}$ denotes users located at region j with pick-up preferences in (5). In this multi-region scenario, users of type j face the constrained logit model (4) where the capacities (4b) are reduced by the vehicles taken by the other users. The resulting distributions $p \coloneqq (p^j)_{j\in\mathcal{R}}$ are thus characterized by

$$\forall j \in \mathcal{R}: \begin{cases} \max\limits_{p^j \in \mathbb{S}} & \overbrace{(p^j)^\top r^j(\theta) - (p^j)^\top \log(p^j)}^{\coloneqq J^j(p^j,\theta)} & \text{(6a)} \\ \text{s.t.} & \sum\limits_{k\in\mathcal{R}} d^{\mathrm{pu},k}\,\Xi\,p^k \le x/\Delta t & \text{(6b)} \end{cases}$$

This collection of logit models coupled through the resource constraints (6b) constitutes a generalized game [23], whose solutions are known as generalized Nash equilibria (GNEs).

The game in (6) admits infinitely many GNEs, which differ in how the shortage of vehicles in each region is split among the user types, technically, in the dual variables of the coupling constraints (6b). Here, we consider the so-called variational GNEs (v-GNEs), in which all user types share the same dual variables for (6b). This models users who decide simultaneously, with no user type having priority in accessing the available vehicles [23].

Remark 1 (Choice of equilibrium). An equally plausible alternative to the v-GNE is a first-come, first-served model. There, earlier users have priority, but users arrive in random order within an interval. Each user chooses via the logit model among the vehicles still available, and the response is the expected outcome over all arrival orders. Without availability constraints, this expectation coincides with the multinomial logit [10]. With constraints, characterizing it is still an open question. Hence, it is beyond the scope of this paper, and we leave it to future work.

The v-GNEs are the solutions to the following variational inequality, or generalized equation, parametric in θ and x:

$$0 \in F(p,\theta) + \mathcal{N}_{\mathcal{Z}(x)}(p), \tag{7}$$

where $\mathcal{N}_{\mathcal{Z}(x)}$ is the normal cone of the feasible set of (6),

$$\mathcal{Z}(x) \coloneqq \left\{p \in \mathbb{S}^N \;\middle|\; \textstyle\sum_{k\in\mathcal{R}} d^{\mathrm{pu},k}\,\Xi\,p^k \le x/\Delta t\right\}, \tag{8}$$

with $\mathbb{S}^N \coloneqq \mathbb{S}\times\cdots\times\mathbb{S}$, and $F(\cdot,\theta) = -\left(\nabla_{p^j} J^j(p^j,\theta)\right)_{j\in\mathcal{R}}$ is the pseudo-gradient of the game in (6), which reads as

$$F(p,\theta) = \log(p) + \mathbb{1} - r(\theta), \tag{9}$$

with $\log(\cdot)$ entry-wise and $r(\theta) \coloneqq (r^j(\theta))_{j\in\mathcal{R}}$.

Since the objectives in (6) are decoupled, the v-GNEs are the maximizers of the social welfare $W(p,\theta) \coloneqq \sum_{j\in\mathcal{R}} J^j(p^j,\theta)$ over $\mathcal{Z}(x)$ [24]. These are unique for any $x \in \mathbb{R}^N_+$ and $\theta \in [0,1]^N$ (see Thm. 2 in Appendix VI-A), yielding the single-valued parameters-to-vGNE mapping

$$p^*(x,\theta) = \left\{p \in \mathbb{S}^N \mid p \text{ solves (7)}\right\}. \tag{10}$$

C. Incentive-Response Map (IRM)

The connection between incentives θ and pick-up flows u^{pu}, given vehicle availability x, is determined by the incentive response mapping (IRM):

$$\mathcal{S}(x,\theta) = D\,p^*(x,\theta), \tag{11}$$

where the matrix $D \coloneqq (d^{\mathrm{pu}})^\top \otimes I_{|\mathcal{C}|}$ weighs the distributions of each user type j by the corresponding demand flow $d^{\mathrm{pu},j}$. The first N entries of $\mathcal{S}(x,\theta)$ are the pick-up flows in (2), i.e., $u^{\mathrm{pu}} = \Xi\,\mathcal{S}(x,\theta)$, and the last is the flow of users leaving the service, $u^{\mathrm{lu}} = \Xi_\emptyset\,\mathcal{S}(x,\theta)$, with $\Xi_\emptyset \coloneqq \begin{bmatrix}0_{1\times N} & 1\end{bmatrix}$.

III. Incentive Design Algorithm

In this section, we formalize the optimal incentive design problem for fleet rebalancing as a mathematical program with nested equilibrium constraints. Then, we develop an efficient first-order algorithm that uses backpropagation and differentiates through the implicit incentive-response mappings.

[3] this depends from the distance between the centers of regions j and i.

[4] If capacity constraints are inactive, the solution of (4) is the multinomial logit [10], commonly used to model vehicle-sharing users [9], [11], [12].

[5] heterogeneous preferences (4a) arise since incentives θ^i are regional, and the base costs w^j depend on the distances from region center j.

A. Optimal control via incentives

Fleet rebalancing can be cast as an optimal control problem in which the incentives are the decision variables and the input of the dynamics (2) is given by the IRM (11). Formally, we aim to find a sequence of incentives $\boldsymbol{\theta} \coloneqq (\theta_0, \dots, \theta_{T-1})$ over a planning horizon $\mathbb{T} \coloneqq \{0, \dots, T-1\}$ that solves

$$\min_{\boldsymbol{\theta}} \quad \sum_{t=0}^{T-1} \ell\left(u_t, \theta_t\right) + h(x_T) \tag{12a}$$

$$\text{s.t.} \quad x_{t+1} = x_t + \Delta t \left(d_t^{\mathrm{do}} - \Xi\, u_t\right), \quad \forall t \in \mathbb{T}, \tag{12b}$$

$$u_t = \mathcal{S}\left(x_t, \theta_t\right), \quad \forall t \in \mathbb{T}, \tag{12c}$$

$$\theta_t \in \Theta, \quad \forall t \in \mathbb{T}, \tag{12d}$$

where x_0 is the initial vehicle distribution, and $\Theta \coloneqq [0, \bar{\theta}]^N$, with $\bar{\theta}$ being the maximum incentive rate; the stage cost

$$\ell(u_t, \theta_t) \coloneqq \gamma_R \|\theta_t\|_2^2 + \gamma_Q \left(\Delta t\, \Xi_\emptyset\, u_t\right)^2 \tag{13}$$

trades off the cost of giving incentives against the number of users leaving the service, with $\gamma_R, \gamma_Q \in \mathbb{R}_+$, while the terminal cost $h(x_T) \coloneqq \gamma_h \|x_T - \bar{x}\|_2^2$ penalizes the deviation from a desired final distribution $\bar{x}$ of the fleet, with $\gamma_h \in \mathbb{R}_+$.

The constraints (12c) involve the IRM, which is nonlinear, nonsmooth and has no explicit expression, and apply it recursively over the horizon through the dynamics (12b), creating a chain of nested equilibrium constraints.

To approach (12), we first recast it in a more compact form. Substituting (12c) into the dynamics (12b) and into the stage cost (13) yields the state update map and the stage cost as functions of the state and the incentives only,

$$\mathcal{T}(x_t, \theta_t) \coloneqq x_t + \Delta t \left(d_t^{\mathrm{do}} - \Xi\, \mathcal{S}(x_t, \theta_t)\right), \tag{14}$$

$$g(x_t, \theta_t) \coloneqq \ell\left(\mathcal{S}(x_t, \theta_t), \theta_t\right). \tag{15}$$

Substituting, in turn, the dynamics into the cost yields

$$\min_{\boldsymbol{\theta}} \quad V_0\left(x_0, \boldsymbol{\theta}\right) \tag{16a}$$

$$\text{s.t.} \quad \boldsymbol{\theta} \in \Omega \coloneqq \Theta \times \cdots \times \Theta. \tag{16b}$$

in which the constraints (12b)–(12c) are embedded in the cost-to-go V_0, defined recursively, for all $t \in \mathbb{T}$, as

$$V_t(x_t, \boldsymbol{\theta}) \coloneqq g(x_t, \theta_t) + V_{t+1}\left(\mathcal{T}(x_t, \theta_t), \boldsymbol{\theta}\right), \tag{17}$$

with $V_T(x_T, \boldsymbol{\theta}) \coloneqq h(x_T)$; namely, $V_t(x_t, \boldsymbol{\theta})$ is the cost accumulated from interval t to the end of the horizon, starting from x_t under the incentives $\boldsymbol{\theta}$. Unrolling (17) from $t = 0$ then yields

$$V_0(x_0, \boldsymbol{\theta}) = \sum_{t=0}^{T-1} g(x_t, \theta_t) + h(x_T), \tag{18}$$

where, for all $t \geq 1$,

$$x_t = \mathcal{T}\Big(\cdots \mathcal{T}\big(\mathcal{T}(x_0, \theta_0), \theta_1\big) \cdots, \theta_{t-1}\Big), \tag{19}$$

which reveals the nested structure of (16).

B. Algorithm design and analysis

To solve (16), we use the first-order scheme for nonsmooth functions summarized in Alg. 1. Computing a "gradient" of V_0 is complicated (i) by the IRM (11), which is implicit and nonsmooth, and (ii) by its nested application along the horizon in (12). To solve these challenges, we adapt the algorithm in [17] to our game-theoretic setup with nonlinear cost functions. Namely, we address (i) by relying on conservative Jacobians (Def. 1), computed via nonsmooth implicit differentiation [22], [25], and (ii) by propagating these Jacobians backward in time along the recursion (17) (Alg. 2–3). The remainder of this section presents these algorithms and establishes convergence guarantees (Thm. 1).

Algorithm 1 Projected conservative-gradient update

Require: x_0, $\boldsymbol{\theta}^{(0)}$, $\{\alpha^{(k)}\}_{k \in \mathbb{N}}$, ε

1: $k \leftarrow 0$
2: **repeat**
3: $\quad J_{V,\boldsymbol{\theta}}^{(k)} \in \mathcal{J}_{\boldsymbol{\theta}} V_0(x_0, \boldsymbol{\theta}^{(k)})$ // computed via Alg. 2 $(x_0, \boldsymbol{\theta}^{(k)})$
4: $\quad \boldsymbol{\theta}^{(k+1)} \leftarrow \Pi_\Omega\big(\boldsymbol{\theta}^{(k)} - \alpha^{(k)} J_{V,\boldsymbol{\theta}}^{(k)}\big)$ // Projected gradient step
5: $\quad k \leftarrow k + 1$
6: **until** $\|\boldsymbol{\theta}^{(k)} - \boldsymbol{\theta}^{(k-1)}\|_2^2 < \varepsilon$
7: **return** $\boldsymbol{\theta}^{(k)}$

To compute a conservative gradient $J_{V,\boldsymbol{\theta}}^{(k)} \in \mathcal{J}_{\boldsymbol{\theta}} V_0(x_0, \boldsymbol{\theta}^{(k)})$ in line 3 of Alg. 1, we leverage the recursive structure of the cost-to-go in (17). Applying the chain rule of differentiation for conservative Jacobians [26, Lemma 5] to (17) expresses $J_{V,x}^{(t)} \in \mathcal{J}_x V_t(x_t, \boldsymbol{\theta})$ and $J_{V,\boldsymbol{\theta}}^{(t)} \in \mathcal{J}_{\boldsymbol{\theta}} V_t(x_t, \boldsymbol{\theta})$ of the cost-to-go at t in terms of the partial Jacobians at $t+1$, and of the partial Jacobians of the stage cost g in (15) and of the state update map $\mathcal{T}$ in (14) at (x_t, θ_t) (lines 6–7). Alg. 2 evaluates this recursion backwards in time from $t = T$, where $V_T(x_T, \boldsymbol{\theta}) = h(x_T)$ by definition (line 2), down to $t = 0$, where $J_{V,\boldsymbol{\theta}}^{(0)}$ is the sought conservative gradient of V_0.

Implementing this recursion requires (i) the state trajectory $\boldsymbol{x}^{(k)} = (x_0, x_1^{(k)}, \dots, x_T^{(k)})$ generated by the control sequence $\boldsymbol{\theta}^{(k)}$ from the initial state x_0, computed by the so-called forward pass in Alg. 3 (line 1), as well as (ii) the the partial Jacobians of the state update map (14) and of the stage cost (15) at (x_t, θ_t). Both depend on (x_t, θ_t) through the IRM, i.e., through the v-GNE mapping p^* in (10). Given elements $J_{p^*,x}^{(t)} \in \mathcal{J}_x p^*(x_t, \theta_t)$ and $J_{p^*,\theta}^{(t)} \in \mathcal{J}_\theta p^*(x_t, \theta_t)$, the chain rule [26, Lemma 5] applied to (11) and (13)–(15) yields

$$J_{g,x}^{(t)} = c_t\, J_{p^*,x}^{(t)}, \tag{20a}$$

$$J_{g,\boldsymbol{\theta}}^{(t)} = e_t^\top \otimes \big(2\gamma_R\, \theta_t^\top + c_t\, J_{p^*,\theta}^{(t)}\big), \tag{20b}$$

and

$$J_{\mathcal{T},x}^{(t)} = I - \Delta t\, \Xi D\, J_{p^*,x}^{(t)}, \tag{21a}$$

$$J_{\mathcal{T},\boldsymbol{\theta}}^{(t)} = e_t^\top \otimes \big(-\Delta t\, \Xi D\, J_{p^*,\theta}^{(t)}\big), \tag{21b}$$

where $c_t \coloneqq 2\gamma_Q \Delta t^2\, u_t^{\mathrm{lu}}\, \Xi_\emptyset D$, with $u_t^{\mathrm{lu}} = \Xi_\emptyset D\, p^*(x_t, \theta_t)$, is the Jacobian of the leaving-user term in (13) with respect to

Algorithm 2 Backpropagation

Require: x_0, $\boldsymbol{\theta} = (\theta_0, \dots, \theta_{T-1})$

$\boldsymbol{x} \leftarrow$ state trajectory of (14) under $\boldsymbol{\theta}$ // via Alg. 3 $(x_0, \boldsymbol{\theta})$

$J_{V,x}^{(T)} \leftarrow 2\gamma_h (x_T - \bar{x})^\top$, $J_{V,\boldsymbol{\theta}}^{(T)} \leftarrow 0$ // cons. Jac. of $V_T = h$

for $t = T-1, \dots, 0$ **do**

  $(J_{g,x}^{(t)}, J_{g,\boldsymbol{\theta}}^{(t)}) \in \mathcal{J} g(x_t, \theta_t)$ // computed via (20)

  $(J_{\mathcal{T},x}^{(t)}, J_{\mathcal{T},\boldsymbol{\theta}}^{(t)}) \in \mathcal{J} \mathcal{T}(x_t, \theta_t)$ // computed via (21)

  $J_{V,x}^{(t)} \leftarrow J_{g,x}^{(t)} + J_{V,x}^{(t+1)} J_{\mathcal{T},x}^{(t)}$ // cons. Jac. of V_t in x_t

  $J_{V,\boldsymbol{\theta}}^{(t)} \leftarrow J_{g,\boldsymbol{\theta}}^{(t)} + J_{V,x}^{(t+1)} J_{\mathcal{T},\boldsymbol{\theta}}^{(t)} + J_{V,\boldsymbol{\theta}}^{(t+1)}$ // cons. Jac. of V_t in $\boldsymbol{\theta}$

end for

return $J_{V,\boldsymbol{\theta}}^{(0)}$ // a conservative gradient of $V_0(x_0, \boldsymbol{\theta})$

Algorithm 3 Forward pass

Require: x_0, $\boldsymbol{\theta} = (\theta_0, \dots, \theta_{T-1})$

for $t = 0, \dots, T-1$ **do**

  $p_t^* \leftarrow$ solution of (7) at x_t, θ_t // v-GNE of (6)

  $x_{t+1} \leftarrow x_t + \Delta t \left(d_t^{\mathrm{do}} - \Xi D\, p_t^*\right)$ // state update (14)

end for

return $\boldsymbol{x} = (x_0, \dots, x_T)$ // state trajectory from x_0 under $\boldsymbol{\theta}$

p. Since $\mathcal{T}$ and g at interval t depend on $\boldsymbol{\theta}$ only through θ_t, their Jacobians with respect to $\boldsymbol{\theta}$ vanish outside the block of θ_t, which we encode via $e_t \in \mathbb{R}^T$.

It remains to compute $J_{p^*,x}^{(t)}$ and $J_{p^*,\theta}^{(t)}$. Since p^* is defined only implicitly, we use the nonsmooth implicit function theorem (IFT) [22]. Applied directly to (7), e.g., via the fixed-point form $p = \Pi_{\mathcal{Z}(x)}\big(p - F(p,\theta)\big)$, the IFT would require a conservative Jacobian of the projection onto $\mathcal{Z}(x)$, which depends on x and has no closed form. Instead, we adapt the dual route of [17] from quadratic to entropic costs. This route avoids differentiating the projection onto $\mathcal{Z}(x)$ by working with the dual problem in the multipliers $\mu = (\mu_{\mathrm{eq}}, \mu_{\mathrm{iq}})$ of the equality and inequality constraints of $\mathcal{Z}(x)$. Its feasible set $\mathcal{M} := \mathbb{R}^N \times \mathbb{R}^N_{\geq 0}$ is a box constraint on the dual variables, independent of x and θ, so the corresponding projection reduces to the entry-wise $\Pi_{\mathcal{M}}(\mu) = \big(\mu_{\mathrm{eq}}, \max(0, \mu_{\mathrm{iq}})\big)$, whose conservative Jacobian is explicit. Appendix VI-B details the derivation and summarizes the procedure in Alg. 4.

Next, we show that algorithm 1 converge to a critical point of (16). This guarantee does not follow directly from [17], whose analysis covers nested optimization problems with quadratic costs. In contrast, (12) involves nested v-GNEs, whose equivalent social-welfare problems have entropic, hence non-quadratic, costs. Nevertheless, the analysis of [17] can be adapted to our setting by exploiting the path-differentiability of the vGNE mapping p^*.

Theorem 1. Let $\boldsymbol{\theta}^{(0)} \in \Omega$, $x_0 \in \mathbb{R}^N_{>0}$, and $d_t^{\mathrm{pu}} \neq 0$ for all $t \in \mathbb{T}$, and let the sequence of step sizes $\{\alpha^{(k)}\}_{k\in\mathbb{N}}$ satisfy: $\alpha^{(k)} > 0$ for all $k \in \mathbb{N}$, $\sum_{k=0}^{\infty} \alpha^{(k)} = \infty$, and $\sum_{k=0}^{\infty} (\alpha^{(k)})^2 < \infty$. Then, the sequence $\{\boldsymbol{\theta}^{(k)}\}_{k\in\mathbb{N}}$ generated by Algorithm 1 converges to a critical point of (16), i.e., a point $\tilde{\boldsymbol{\theta}} \in \Omega$ such that $0 \in \mathcal{J}_{\boldsymbol{\theta}} V_0(x_0, \tilde{\boldsymbol{\theta}}) + \mathcal{N}_\Omega(\tilde{\boldsymbol{\theta}})$. □

The proof follows arguments analogous to [17], adapted to lower-level problems with entropic cost functions as in (7), and is omitted due to space limitations. For completeness, we provide it in the extended version of this paper [27, App. C].

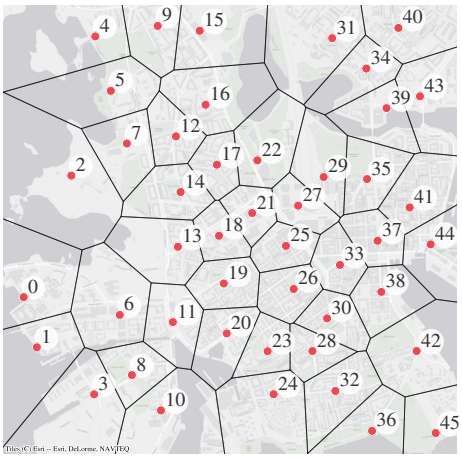


Fig. 1. The free-floating bicycle sharing system in central Helsinki restricted to $N = 46$ regions. Each region is associated with a center indicated by a red dot and an index number. The black lines show boundaries between regions. Distances between neighboring region centers vary between 278 and 2119 m, with an average of 604 m. Basemap: Esri World Light Gray Canvas (Tiles © Esri — Esri, DeLorme, NAVTEQ)

Daniel: Reminder! Update arXiv citation

IV. Numerical results

We evaluate our approach on the Helsinki Region Transport (HSL) city bicycle dataset [21], restricted to $N = 46$ regions (see Figure 1) and discretized into $T = 96$, 15-minute time steps over a representative day (May 1, 2025). We set the incentive sensitivity uniformly ($\lambda^{j,i} = 500$) for all region pairs, meaning a full incentive ($\theta^i = 1$) compensates 500 m of walking distance. According to [4], most users are unwilling to walk farther than this to book an alternative vehicle. The distances between neighboring regions in our dataset vary between 278 and 2119 m, with an average of 604 m; hence, a sufficiently high incentive can draw users to a nearby region in many cases. We set $w^{j,\emptyset} = 100$, i.e., users tolerate walking up to 100 m to a neighboring station without any incentive before leaving the service. Since the smallest distance between neighboring regions (278 m) exceeds this tolerance, users always prefer to leave rather than switch regions absent incentives, reflecting an unforgiving baseline behavior. Finally, we set the entropy regularization temperature ($\tau = 50$) to sharpen decision boundaries in the logit model. We compute the v-GNEs with CVXPY [28] using the SCS solver [29]. The code and data to reproduce all numerical results are publicly available.[6]

[6] https://github.com/tihanyidani/vehicle-sharing

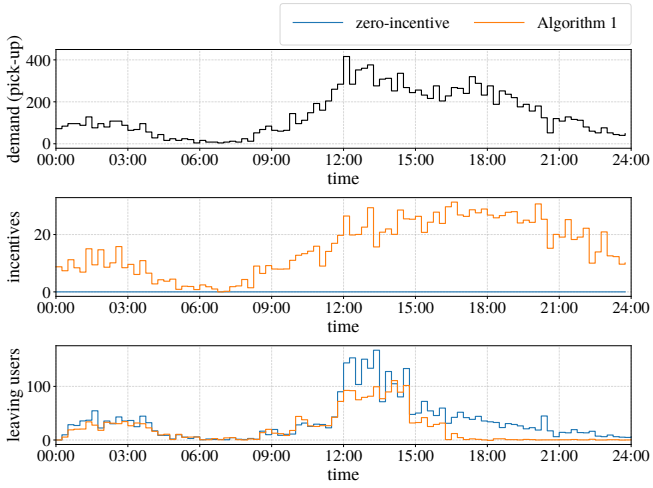


Fig. 2. Incentive trajectory optimized by Algorithm 1 against the zero-incentive baseline. All plots show aggregated flow across all regions against the time of day. Top: aggregated user demand flow for picking up vehicles, $\sum_{i\in\mathcal{R}} d_t^{\mathrm{pu},i}$. Center: aggregated incentive trajectories, $\sum_{i\in\mathcal{R}} \theta_t^i$, generated by Algorithm 1 (orange). Bottom: flow of users leaving the service, u_t^{lu}, without incentives (blue) and with incentives generated by Algorithm 1 (orange).

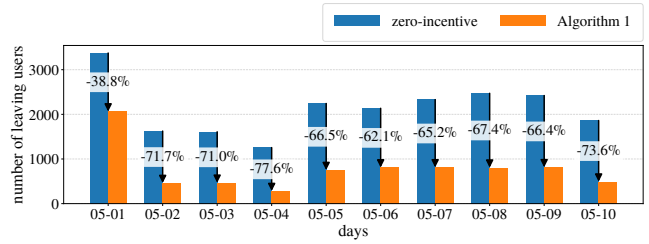


Fig. 3.

A. Reduction in leaving users

We show that applying our optimized incentive trajectory reduces the total number of leaving users by 38%. This improvement follows directly from the algorithm's response to demand overshoot: as pickup demand exceeds available vehicles during peak hours (11 a.m.–8 p.m.), the algorithm increases incentives accordingly, reducing the number of users abandoning the service in the afternoon. This effect is consistent across 10 different days in the dataset, with reductions ranging from 38% to 73% (see Figure 3).

We compare the performance of our approach in Algorithm 1 against the zero-incentive baseline and three naive alternatives: i) Trigger, a threshold-triggered policy, where $\theta_t^i = \bar{\theta}$, if there are more vehicles in region i than threshold $P_{\mathrm{trig.}}$, $x_t^i \geq P_{\mathrm{trig.}}$, and $\theta_t^i = 0$ otherwise; ii) Proportional, a proportional-error policy, where $\theta_t^i = P_{\mathrm{prop.}} \cdot (x_t^i - d_t^{\mathrm{pu},i})$, the incentives are set proportional to the overshoot of available vehicles over the demanded vehicles; iii) One-step, a short-horizon policy, where θ_t is generated by solving (12) for a single-step horizon $T = 1$. We summarized the results in Table I. All four methods achieve similar reductions in leaving users (32.9%–38.8%). However, Algorithm 1 uses substantially less incentive budget to reach a comparable outcome, yielding more than $1.2\times$ to $2.0\times$ the reduction achieved per unit of incentive.

Algorithms	Leaving users	Reduction	Incentive usage	Incentive efficiency
zero-incentive	3374	-	-	-
Algorithm 1	2064	-38.8%	1460	-0.027%
trigger	2265	-32.9%	2176	-0.015%
proportional	2172	-35.6%	2728	-0.013%
one-step	2087	-38.1%	1791	-0.021%

TABLE I. Comparison of the five applied algorithms: i) Leaving users – the total number of leaving users throughout the day across all regions; ii) Reduction – improvement in the reduction in leaving user numbers compared to the zero-incentive baseline; iii) Incentive usage – total amount of incentives broadcast throughout the day across all regions; iv) Incentive efficiency – reduction divided by incentive usage.

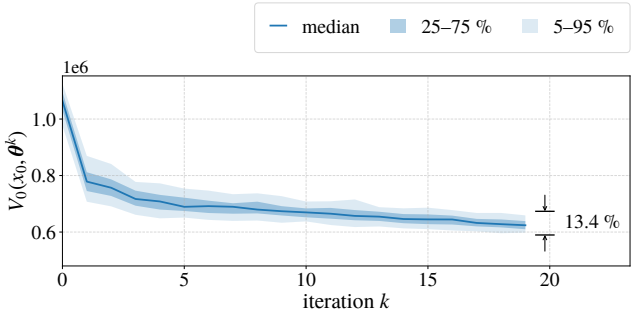


Fig. 4. Different initial $\boldsymbol{\theta}^{(0)}$.

B. Convergence behaviour

Since (16) is non-convex, different initial guesses may converge to different critical points. To assess the practical impact of this, we ran 100 experiments with different uniformly randomized initial guesses $\boldsymbol{\theta}^{(0)}$ and tracked the cost function value across iterations for each run (Figure 4). Across all 100 randomized runs, the algorithm shows the same convergence behavior. Comparing the converged solutions directly, we find that the final costs are clustered across the 100 runs, with the maximal difference in cost between any two solutions of 13.4%.

V. Conclusion

We addressed optimal incentive design for user-based rebalancing in vehicle-sharing systems, modeling users' responses as a variational GNE of availability-constrained logit models. The resulting problem with nested equilibrium constraints was solved via projected gradient descent with nonsmooth implicit differentiation, with guaranteed convergence to critical points. On Helsinki bike-sharing data, the scheme reduced leaving users by 38–73% relative to no incentives, using less budget than heuristic and myopic policies. Future work includes a receding-horizon implementation robust to uncertain demand, combination with operator-based relocation, drop-off incentives, and modeling users' responses via the first-come, first-served model of Remark 1.

VI. Appendix

A. Existence and uniqueness of the v-GNE

Theorem 2. For any $x \in \mathbb{R}^N_+$, $\theta \in [0,1]^N$, the generalized equation in (7) admits a unique solution, $p^*(x,\theta)$. □

Proof. Existence. For any $x \in \mathbb{R}^N_+$, $\mathcal{Z}(x)$ is compact and nonempty, since $\hat{p} \in \mathbb{S}^N$ with all users leaving is always feasible: $\forall j \in \mathcal{R}$, $\hat{p}^{j,\emptyset} = 1$ and $\hat{p}^{j,i} = 0$ if $i \neq \emptyset$ satisfies $\sum_{k\in\mathcal{R}} d^{\mathrm{pu},k}\, \Xi\, \hat{p}^k \leq x/\Delta t$ for any $x \in \mathbb{R}^N_+$. Consequently, the maximizer exists. *Uniqueness.* The Hessian of the social welfare is $\nabla^2_p W(p,\theta) = -\mathrm{diag}(p)^{-1} \prec 0$ for all $p > 0$. Thus, $W(\cdot,\theta)$ is strictly concave on the interior of $\mathbb{S}^N$, and, by continuity of $t \mapsto t\log t$ on $[0,\infty)$ with $0\log 0 := 0$, on all of $\mathbb{S}^N$. Therefore, the maximizer is unique. □

B. Conservative Jacobians of the v-GNE

This appendix details the computation of the conservative Jacobians $J_{p^*,x} \in \mathcal{J}_x p^*(x,\theta)$ and $J_{p^*,\theta} \in \mathcal{J}_\theta p^*(x,\theta)$ of the v-GNE mapping (10), required in (20)–(21). The procedure is summarized in Alg. 4; we omit the time index t throughout.

Algorithm 4 Conservative Jacobians of the v-GNE

Require: x, θ

Step 1: Solving the v-GNE and its dual

1: $p^*, \mu^* \leftarrow$ solution of (7) at x,θ // Solve (7) and its dual

2: $P^* \leftarrow \mathrm{diag}(p^*)$ // Shorthand

Step 2: Differentiating the projection onto $\mathcal{M}$

3: $\mu' = (\mu'_{\mathrm{eq}}, \mu'_{\mathrm{iq}}) \leftarrow \mu^* + Gp^* - b(x)$ // Argument in (23)

4: $\Sigma \leftarrow \mathrm{diag}(\mathbb{1}_N, \sigma(\mu'_{\mathrm{iq}}))$ // Diff. $\Pi_{\mathcal{M}}$, see (24)

Step 3: Implicit differentiation of the dual solution

5: $J_{\mathcal{F},\mu} \leftarrow \Sigma(I - GP^*G^\top) - I$ // Diff. (23) in μ

6: $J_{\mathcal{F},x} \leftarrow -\frac{1}{\Delta t}\Sigma E, \quad J_{\mathcal{F},\theta} \leftarrow \Sigma G P^* \Lambda$ // Diff. (23)

7: $J_{\mu^*,x} \leftarrow -J^{-1}_{\mathcal{F},\mu} J_{\mathcal{F},x}$ // IFT [22, Thm. 5]

8: $J_{\mu^*,\theta} \leftarrow -J^{-1}_{\mathcal{F},\mu} J_{\mathcal{F},\theta}$ // IFT [22, Thm. 5]

Step 4: Differentiating the primal solution

9: $J_{p^*,x} \leftarrow -P^*G^\top J_{\mu^*,x}$ // Diff. (22)

10: $J_{p^*,\theta} \leftarrow P^*(\Lambda - G^\top J_{\mu^*,\theta})$ // Diff. (22)

11: **return** $J_{p^*,x}, J_{p^*,\theta}$

Step 1 (Dual of the social welfare maximization). We write the constraints of $\mathcal{Z}(x)$ in (8) compactly as $G_{\mathrm{eq}}\, p = \mathbb{1}_N$ and $G_{\mathrm{iq}}\, p \leq x/\Delta t$, with $G_{\mathrm{eq}} := I_N \otimes \mathbb{1}^\top_{|\mathcal{C}|}$ and $G_{\mathrm{iq}} := \Xi\, D$, and stack them as $G := (G_{\mathrm{eq}}, G_{\mathrm{iq}})$ and $b(x) := (\mathbb{1}_N, x/\Delta t)$. Let $\mu \in \mathcal{M}$ be the corresponding multipliers. Since the v-GNE maximizes the social welfare $W(\cdot,\theta)$ over $\mathcal{Z}(x)$ (Section II-B), stationarity of the Lagrangian yields

$$p^*(x,\theta) = p(\mu^*,\theta) := \exp\left(r(\theta) - \mathbb{1} - G^\top \mu^*\right), \qquad (22)$$

where $r(\theta) = \Lambda\theta - w$, with $\Lambda := (\Lambda^j)_{j\in\mathcal{R}}$, $\Lambda^j := \big(\mathrm{diag}(\lambda^j), 0_{1\times N}\big)$, and $w := (w^j)_{j\in\mathcal{R}}$, and where $\mu^* = \mu^*(x,\theta)$ minimizes the dual function $\mathbb{1}^\top p(\mu,\theta) + b(x)^\top \mu$ over $\mathcal{M}$. In contrast to the quadratic case of [17], where this primal-dual map is affine, here it is an entry-wise exponential. By (22), $p^* > 0$, hence $P^* := \mathrm{diag}(p^*) \succ 0$. Moreover, G has full row rank whenever $d^{\mathrm{pu}} \neq 0$: the $\emptyset$-entry of each block j appears only in the row of G_{eq} associated with type j, and the rows of G_{iq} are nonzero and have disjoint supports. Hence, the linear independence constraint qualification (LICQ) holds and μ^* is unique.

Step 2 (Fixed-point characterization). Since μ^* minimizes the dual function over $\mathcal{M}$, it is a fixed point of a projected gradient step on it, i.e.,

$$0 = \mathcal{F}(\mu^*, x, \theta) := \Pi_{\mathcal{M}}\big(\mu^* + G\, p(\mu^*,\theta) - b(x)\big) - \mu^*, \quad (23)$$

where the parameters enter $\mathcal{F}$ only through $b(x)$ and $r(\theta)$. The map $\mathcal{F}$ is a composition of affine maps, the entry-wise exponential, and $\max(0,\cdot)$; hence, it is locally Lipschitz and definable in the o-minimal structure $\mathbb{R}_{\exp}$, and thus path-differentiable [22]. Its only nonsmooth component is $\Pi_{\mathcal{M}}$, evaluated at μ' in line 3. An element of its conservative Jacobian is $\Sigma \in \mathcal{J}_\mu \Pi_{\mathcal{M}}(\mu')$ in line 4, where $\sigma(\cdot)$ is the step function applied entry-wise to vectors,

$$\sigma(v) := \begin{cases} 1, & \text{if} \quad v > 0, \\ 0, & \text{if} \quad v \leq 0. \end{cases} \qquad (24)$$

Step 3 (Implicit differentiation of the dual solution). By (22), the Jacobians of $p(\mu,\theta)$ at (μ^*,θ) are $-P^*G^\top$ with respect to μ and $P^*\Lambda$ with respect to θ. Hence, differentiating (23) via the chain rule [26, Lemma 5] yields the elements $J_{\mathcal{F},\mu} \in \mathcal{J}_\mu \mathcal{F}$, $J_{\mathcal{F},x} \in \mathcal{J}_x \mathcal{F}$, and $J_{\mathcal{F},\theta} \in \mathcal{J}_\theta \mathcal{F}$ at (μ^*,x,θ) in lines 5–6, where $E := (0_{N\times N}, I_N)$ is the Jacobian of $b(x)$ scaled by Δt. Since G has full row rank and $P^* \succ 0$, $J_{\mathcal{F},\mu}$ is invertible by an argument analogous to [17]. Hence, by the IFT [22, Thm. 5], μ^* is path-differentiable, with the conservative Jacobians in lines 7–8. Note that $J_{\mathcal{F},\mu}$ has size $2N$, rather than $N|\mathcal{C}|$ as in the primal route.

Step 4 (Conservative Jacobians of the primal solution). Since p^* is an explicit function of μ^* and θ by (22), the chain rule maps the dual Jacobians to $J_{p^*,x} \in \mathcal{J}_x p^*(x,\theta)$ and $J_{p^*,\theta} \in \mathcal{J}_\theta p^*(x,\theta)$ in lines 9–10, which are then used in (20)–(21).

C. Proof of Theorem 1

The proof adapts the analysis of [17], developed for nested optimization problems with quadratic lower levels, to our nested v-GNEs, whose equivalent social-welfare problems have entropic, hence non-quadratic, costs. We first recall the tools from nonsmooth analysis used in the proof, then establish the regularity of the v-GNE map p^* through the dual characterization used in Algorithm 4, and finally prove Theorem 1.

a) Preliminaries:

Lemma 1 ([26, Prop. 2])**.** Locally Lipschitz functions that are definable in an o-minimal structure are path-differentiable.

Lemma 2 (Chain rule [26, Lemma 5])**.** Let $\psi_1 : \mathbb{R}^n \to \mathbb{R}^m$ and $\psi_2 : \mathbb{R}^m \to \mathbb{R}^l$ be path-differentiable with conservative Jacobians $\mathcal{J}\psi_1$ and $\mathcal{J}\psi_2$. Then $\psi_2 \circ \psi_1$ is path-differentiable, and $\{AB \mid A \in \mathcal{J}\psi_2(\psi_1(z)),\ B \in \mathcal{J}\psi_1(z)\}$ is a conservative Jacobian for it at z.

Since sums and stackings of path-differentiable maps are path-differentiable, with conservative Jacobians given by the sums and stackings of the individual ones [26], we also apply Lemma 2 to maps of several arguments.

Theorem 3 (Implicit function theorem [22, Thm. 5])**.** Let $f : \mathbb{R}^m \times \mathbb{R}^n \to \mathbb{R}^m$ be path-differentiable with conservative Jacobian $\mathcal{J}f$, and let $f(\tilde{z}, \tilde{\eta}) = 0$. Assume that U is invertible for every $[U \;\; V] \in \mathcal{J}f(\tilde{z}, \tilde{\eta})$, with $U \in \mathbb{R}^{m\times m}$. Then there exist open neighborhoods $\mathcal{U}$ of $\tilde{z}$ and $\mathcal{V}$ of $\tilde{\eta}$, and a path-differentiable map $z^* : \mathcal{V} \to \mathcal{U}$, such that $f(z^*(\eta), \eta) = 0$ for all $\eta \in \mathcal{V}$, and

$$\mathcal{J}z^*(\eta) \coloneqq \big\{ -U^{-1}V \;\big|\; [U \;\; V] \in \mathcal{J}f(z^*(\eta), \eta)\big\} \tag{25}$$

is a conservative Jacobian for z^*.

Theorem 4 ([30, Thm. 6.2])**.** Let $f : \mathbb{R}^n \to \mathbb{R}$ be path-differentiable with conservative Jacobian $\mathcal{J}f$, and let $\mathcal{K} \subset \mathbb{R}^n$ be a nonempty polytope. Consider the iteration

$$z^{(k+1)} = \Pi_{\mathcal{K}}\big(z^{(k)} - \alpha^{(k)} J^{(k)}\big), \quad J^{(k)} \in \mathcal{J}f(z^{(k)}), \tag{26}$$

with $z^{(0)} \in \mathcal{K}$ and step sizes $\{\alpha^{(k)}\}_{k\in\mathbb{N}}$ satisfying the Robbins–Monro conditions in Theorem 1. If $\sup_k \|z^{(k)}\| < \infty$, then $\{z^{(k)}\}_{k\in\mathbb{N}}$ converges to a critical point of f over $\mathcal{K}$, i.e., a point $\tilde{z} \in \mathcal{K}$ such that $0 \in \mathcal{J}f(\tilde{z}) + \mathcal{N}_{\mathcal{K}}(\tilde{z})$.

b) Regularity of the incentive-response map: Throughout this part, we drop the time index and use the notation of Algorithm 4, i.e., $G = (G_{\mathrm{eq}}, G_{\mathrm{iq}})$, $b(x)$, $\mathcal{M}$, $\Pi_{\mathcal{M}}$, E, and $p(\mu, \theta)$ in (22). The dual function of the social welfare maximization over $\mathcal{Z}(x)$ is

$$q(\mu, x, \theta) \coloneqq \mathbb{1}^\top p(\mu, \theta) + b(x)^\top \mu, \tag{27}$$

whose gradient and Hessian with respect to μ read

$$\begin{aligned} \nabla_\mu q &= b(x) - G\, p(\mu, \theta), \\ \nabla^2_{\mu\mu} q &= G \operatorname{diag}\big(p(\mu, \theta)\big)\, G^\top. \end{aligned} \tag{28}$$

Hence, (23) reads $\mathcal{F}(\mu, x, \theta) = \Pi_{\mathcal{M}}\big(\mu - \nabla_\mu q(\mu, x, \theta)\big) - \mu$.

Lemma 3 (LICQ)**.** If $d^{\mathrm{pu}} \neq 0$, then G has full row rank, i.e., the constraints of $\mathcal{Z}(x)$ satisfy the LICQ for any $x \in \mathbb{R}^N$.

Proof. Let $a, c \in \mathbb{R}^N$ satisfy $G_{\mathrm{eq}}^\top a + G_{\mathrm{iq}}^\top c = 0$. The entry of this vector associated with $p^{j,\emptyset}$ is a^j, since $G_{\mathrm{iq}} = \Xi D$ does not involve the leaving options; hence, $a = 0$. The entry associated with $p^{j,i}$, $i \in \mathcal{R}$, is then $d^{\mathrm{pu},j} c^i$. Choosing j with $d^{\mathrm{pu},j} > 0$ yields $c = 0$. □

Lemma 4 (Dual solution)**.** Let $x \in \mathbb{R}^N_{>0}$, $\theta \in \mathbb{R}^N$, and $d^{\mathrm{pu}} \neq 0$. Then $q(\cdot, x, \theta)$ admits a unique minimizer $\mu^*(x, \theta)$ over $\mathcal{M}$, which is the unique solution of $\mathcal{F}(\mu, x, \theta) = 0$, and $p^*(x, \theta) = p(\mu^*(x, \theta), \theta)$.

Proof. Let $\hat{p}$ be given by $\hat{p}^{j,i} = \varepsilon$ for $i \in \mathcal{R}$ and $\hat{p}^{j,\emptyset} = 1 - N\varepsilon$, for all $j \in \mathcal{R}$. Since $x > 0$, for $\varepsilon > 0$ small enough, $\hat{p} > 0$ and $G_{\mathrm{iq}}\hat{p} = \varepsilon(\mathbb{1}^\top d^{\mathrm{pu}})\mathbb{1} < x/\Delta t$. Hence, Slater's condition holds for the (concave, affinely constrained) social welfare maximization, so strong duality holds and the dual minimum over $\mathcal{M}$ is attained. Stationarity of the Lagrangian yields $p^* = p(\mu^*, \theta)$ as in (22). By (22), $p(\mu, \theta) > 0$ for all μ; with Lemma 3, (28) gives $\nabla^2_{\mu\mu} q \succ 0$, so $q(\cdot, x, \theta)$ is strictly convex and its minimizer is unique. Finally, μ minimizes the convex differentiable $q(\cdot, x, \theta)$ over the closed convex set $\mathcal{M}$ if and only if $\mu = \Pi_{\mathcal{M}}(\mu - \nabla_\mu q(\mu, x, \theta))$ [31], i.e., $\mathcal{F}(\mu, x, \theta) = 0$. □

Lemma 5. The map $\mathcal{F}$ in (23) is locally Lipschitz and definable in the o-minimal structure $\mathbb{R}_{\exp}$, hence path-differentiable. Moreover, with $\mu' \coloneqq \mu + Gp(\mu, \theta) - b(x)$ and $P \coloneqq \operatorname{diag}(p(\mu, \theta))$, a conservative Jacobian of $\mathcal{F}$ at (μ, x, θ) is

$$\mathcal{J}\mathcal{F} \coloneqq \big\{[J_{\mathcal{F},\mu} \;\; J_{\mathcal{F},x} \;\; J_{\mathcal{F},\theta}] \;\big|\; \Sigma \in \mathcal{J}\Pi_{\mathcal{M}}(\mu')\big\}, \tag{29}$$

where $J_{\mathcal{F},\mu}$, $J_{\mathcal{F},x}$, $J_{\mathcal{F},\theta}$ are as in lines 5–6 of Algorithm 4, with P^* replaced by P, and $\mathcal{J}\Pi_{\mathcal{M}}(\mu') \coloneqq \{\operatorname{diag}(\mathbb{1}_N, s) \mid s \in [0,1]^N,\ s^i = \sigma(\mu'^{\,i}_{\mathrm{iq}}) \text{ if } \mu'^{\,i}_{\mathrm{iq}} \neq 0\}$. In particular, Σ in line 4 of Algorithm 4 belongs to $\mathcal{J}\Pi_{\mathcal{M}}(\mu')$.

Proof. The map $\Pi_{\mathcal{M}}(\mu) = (\mu_{\mathrm{eq}}, \max(0, \mu_{\mathrm{iq}}))$ is piecewise linear, hence semialgebraic and definable in $\mathbb{R}_{\exp}$ [32]; the entry-wise exponential is definable in $\mathbb{R}_{\exp}$, which is o-minimal by [33]; and $r(\theta) = \Lambda\theta - w$ and $b(x)$ are affine. Since compositions and sums of definable maps are definable [32], and compositions of locally Lipschitz maps are locally Lipschitz, $\mathcal{F}$ is locally Lipschitz and definable, hence path-differentiable by Lemma 1. Since $\Pi_{\mathcal{M}}$ acts entry-wise, the product of the Clarke subdifferentials of its entries, i.e., $\mathcal{J}\Pi_{\mathcal{M}}$, is a conservative Jacobian for it [26]. The Jacobians of the smooth map $(\mu, x, \theta) \mapsto \mu'$ are $I - GPG^\top$, $-\frac{1}{\Delta t}E$, and $GP\Lambda$, by (22). Lemma 2 then yields (29). The last claim follows from (24). □

Lemma 6 (Invertibility of $J_{\mathcal{F},\mu}$)**.** Let $d^{\mathrm{pu}} \neq 0$. For any (μ, x, θ), every $J_{\mathcal{F},\mu}$ such that $[J_{\mathcal{F},\mu} \;\; J_{\mathcal{F},x} \;\; J_{\mathcal{F},\theta}] \in \mathcal{J}\mathcal{F}(\mu, x, \theta)$ in (29) is invertible.

Proof. Let $J_{\mathcal{F},\mu} = \Sigma(I - H) - I$, with $\Sigma = \operatorname{diag}(s)$, $s \in [0,1]^{2N}$, and $H \coloneqq GPG^\top$, and let $J_{\mathcal{F},\mu} v = 0$, i.e., $\Sigma(I - H)v = v$. Let $\mathcal{A} \coloneqq \{k \mid s^k > 0\}$, which contains all indices of μ_{eq}. For $k \notin \mathcal{A}$, the row associated with k gives $v^k = 0$. For $k \in \mathcal{A}$, it gives $(Hv)^k = -\frac{1-s^k}{s^k} v^k$. Since v is supported on $\mathcal{A}$, this reads

$$\big(G^{\mathcal{A}} P (G^{\mathcal{A}})^\top + \Gamma\big) v^{\mathcal{A}} = 0, \quad \Gamma \coloneqq \operatorname{diag}\big(\tfrac{1-s^k}{s^k}\big)_{k\in\mathcal{A}} \succeq 0, \tag{30}$$

where $G^{\mathcal{A}}$ collects the rows of G indexed by $\mathcal{A}$. By Lemma 3, $G^{\mathcal{A}}$ has full row rank, and $P \succ 0$ by (22). Hence, $G^{\mathcal{A}} P (G^{\mathcal{A}})^\top + \Gamma \succ 0$, so $v^{\mathcal{A}} = 0$ and $v = 0$. □

Lemma 6 covers the active ($s^k = 1$), inactive ($s^k = 0$), and weakly active ($s^k \in [0,1]$) constraints of [17] at once, which is required since Theorem 3 asks for invertibility of every element of the conservative Jacobian.

Lemma 7. Let $d^{\mathrm{pu}} \neq 0$. Then μ^* and p^* are path-differentiable on $\mathcal{O} \coloneqq \mathbb{R}^N_{>0} \times \mathbb{R}^N$, and the matrices $J_{\mu^*,x}, J_{\mu^*,\theta}$ and $J_{p^*,x}, J_{p^*,\theta}$ in lines 7–8 and 9–10 of Algorithm 4 are elements of conservative Jacobians of μ^* and p^* at (x, θ).

Proof. Fix $(\tilde{x}, \tilde{\theta}) \in \mathcal{O}$. By Lemma 4, $\mathcal{F}(\mu^*(\tilde{x}, \tilde{\theta}), \tilde{x}, \tilde{\theta}) = 0$. By Lemmas 5 and 6, Theorem 3 applies with $z = \mu$ and $\eta = (x, \theta)$, and yields a path-differentiable z^* with $\mathcal{F}(z^*(x, \theta), x, \theta) = 0$ on a neighborhood of $(\tilde{x}, \tilde{\theta})$, which we take inside the open set $\mathcal{O}$. By the uniqueness in Lemma 4, $z^* = \mu^*$ on this neighborhood. Since path-differentiability is a local property, μ^* is path-differentiable on $\mathcal{O}$, and, by (25) and (29), lines 7–8 return an element of its conservative Jacobian. Finally, $p^* = p(\mu^*, \theta)$ by Lemma 4, with $p(\cdot, \cdot)$ in (22) smooth with Jacobians $-P^* G^\top$ and $P^* \Lambda$; Lemma 2 yields lines 9–10. □

c) Regularity of the cost-to-go: Lemma 7 requires $x \in \mathbb{R}^N_{>0}$; for $x^i = 0$, $p^{*,j,i} = 0$ for all j with $d^{\mathrm{pu},j} > 0$, contradicting $p^* > 0$ in (22), so the dual minimum is not attained. The next lemma ensures that the states remain positive.

Lemma 8. Let $x_0 \in \mathbb{R}^N_{>0}$ and $d_t^{\mathrm{do}} \in \mathbb{R}^N_{>0}$ for all $t \in \mathbb{T}$. Then, for any $\boldsymbol{\theta} \in \mathbb{R}^{TN}$, the states x_t generated by (14) from x_0 under $\boldsymbol{\theta}$ satisfy $x_t \in \mathbb{R}^N_{>0}$ for all $t \in \{0, \dots, T\}$.

Proof. By induction on t. If $x_t > 0$, then $p^*(x_t, \theta_t) \in \mathcal{Z}(x_t)$ gives $\Delta t \, \Xi D p^*(x_t, \theta_t) \leq x_t$, so, by (14), $x_{t+1} = \mathcal{T}(x_t, \theta_t) \geq \Delta t \, d_t^{\mathrm{do}} > 0$. □

Lemma 9. Let the assumptions of Theorem 1 hold, and let $d_t^{\mathrm{do}} \in \mathbb{R}^N_{>0}$ for all $t \in \mathbb{T}$. Then $V_0(x_0, \cdot)$ is path-differentiable on $\mathbb{R}^{TN}$, and, for any $\boldsymbol{\theta} \in \mathbb{R}^{TN}$, the output $J^{(0)}_{V,\boldsymbol{\theta}}$ of Algorithm 2 belongs to a conservative Jacobian $\mathcal{J}_{\boldsymbol{\theta}} V_0(x_0, \boldsymbol{\theta})$.

Proof. By Lemma 7, the maps $\mathcal{T}$ in (14) and g in (15) are path-differentiable on $\mathcal{O}$, as compositions of p^* with polynomial maps (Lemma 2); (20)–(21) follow from Lemma 2 applied to (14)–(15), so $[J^{(t)}_{g,x} \;\; J^{(t)}_{g,\boldsymbol{\theta}}] \in \mathcal{J} g(x_t, \theta_t)$ and $[J^{(t)}_{\mathcal{T},x} \;\; J^{(t)}_{\mathcal{T},\boldsymbol{\theta}}] \in \mathcal{J}\mathcal{T}(x_t, \theta_t)$, viewed as functions of $(x_t, \boldsymbol{\theta})$. Note that $\mathcal{T}(x, \theta_t) \in \mathbb{R}^N_{>0}$ for any $x \in \mathbb{R}^N_{>0}$ by the proof of Lemma 8.

We show by backward induction that, for $t = T, \dots, 0$, V_t is path-differentiable on $\mathbb{R}^N_{>0} \times \mathbb{R}^{TN}$ and $[J^{(t)}_{V,x} \;\; J^{(t)}_{V,\boldsymbol{\theta}}] \in \mathcal{J} V_t(x_t, \boldsymbol{\theta})$. For $t = T$, $V_T = h$ is smooth with gradient as in line 2 of Algorithm 2. For $t < T$, (17) writes V_t as the sum of g and the composition of V_{t+1} with $(x, \boldsymbol{\theta}) \mapsto (\mathcal{T}(x, \theta_t), \boldsymbol{\theta})$, all path-differentiable; Lemma 2 yields exactly lines 6–7. At $t = 0$, with x_0 fixed, $J^{(0)}_{V,\boldsymbol{\theta}} \in \mathcal{J}_{\boldsymbol{\theta}} V_0(x_0, \boldsymbol{\theta})$. Since, by Lemma 8, all states remain in $\mathbb{R}^N_{>0}$ for any $\boldsymbol{\theta} \in \mathbb{R}^{TN}$, $V_0(x_0, \cdot)$ is path-differentiable on $\mathbb{R}^{TN}$. □

d) Proof of Theorem 1: We apply Theorem 4 with $f = V_0(x_0, \cdot)$, $\mathcal{K} = \Omega$, and $z^{(k)} = \boldsymbol{\theta}^{(k)}$. First, f is path-differentiable on $\mathbb{R}^{TN}$ by Lemma 9. Second, $\Omega = [0, \bar{\theta}]^{TN}$ is a nonempty box, hence a polytope. Third, by Lemma 9, $J^{(k)}_{V,\boldsymbol{\theta}}$ in line 3 of Algorithm 1 belongs to $\mathcal{J}_{\boldsymbol{\theta}} V_0(x_0, \boldsymbol{\theta}^{(k)})$, so line 4 is an instance of (26) (with $\varepsilon = 0$, i.e., without termination). Fourth, the step sizes satisfy the Robbins–Monro conditions by assumption. Finally, $\boldsymbol{\theta}^{(k)} \in \Omega$ for all k, and Ω is compact, so $\sup_k \|\boldsymbol{\theta}^{(k)}\| < \infty$. Hence, by Theorem 4, $\{\boldsymbol{\theta}^{(k)}\}_{k \in \mathbb{N}}$ converges to a point $\tilde{\boldsymbol{\theta}} \in \Omega$ with $0 \in \mathcal{J}_{\boldsymbol{\theta}} V_0(x_0, \tilde{\boldsymbol{\theta}}) + \mathcal{N}_\Omega(\tilde{\boldsymbol{\theta}})$, i.e., a critical point of (16), which is (12) with the constraints (12b)–(12c) eliminated. ■

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